

SUDARSHAN UAV — OPERATOR MANUAL

Issue 1.0 | CONFIDENTIAL — Authorized Personnel Only

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1. Introduction & Operating Limitations

The SUDARSHAN is a custom-built quadrotor UAV designed for indoor and outdoor low-altitude research and training operations. It is operated exclusively under direct visual line-of-sight (VLOS) by trained personnel. This manual describes the procedures, limitations, and systems information required for safe and effective operation.

Operating Limitations

Parameter	Limit	Notes
Maximum Wind Speed	5 m/s	Operations shall cease above this value
Maximum Operating Altitude	30 m AGL	Hard limit enforced by operator discipline
Maximum Control Range	50 m (Wi-Fi range)	GCS signal degrades beyond 50 m line-of-sight
Operating Temperature	0 °C to +40 °C	Battery performance degrades below 10 °C
Maximum Payload	150 g	Exceeding payload increases instability
Minimum Battery for Flight	10.5 V (3.5 V/cell)	Do not ARM below this voltage
Maximum Continuous Hover	12 min (2200 mAh)	Land with ≥ 2 min reserve (≈9.9 V)
Propeller Size	8 × 4.5 inch	Replacement must match specification exactly

■ WARNING

SUDARSHAN is not certified for operations over people, moving vehicles, or within 5 km of any aerodrome without explicit ATC authorization. The operator bears full responsibility for compliance with local UAV regulations.

2. System Description

2.1 Configuration

SUDARSHAN is an X-configuration quadrotor. Motors are designated FL (front-left), FR (front-right), RL (rear-left), RR (rear-right). FL and RR rotate clockwise (CW); FR and RL rotate counter-clockwise (CCW). Frame diagonal motor-to-motor distance: approximately 450 mm.

2.2 Weight & Balance

Component	Mass	Notes
Airframe (bare)	≈ 450 g	Including arms and landing gear
Motors × 4	≈ 200 g	2212/920 KV brushless
LiPo Battery	≈ 190 g	3S 2200 mAh 30C
Electronics	≈ 120 g	Mega, Uno, ESP32, PCA9685, sensors
TOTAL (no payload)	≈ 960 g	All-up weight without payload

2.3 Electrical System

A 3S LiPo (11.1 V nominal, 12.6 V fully charged) feeds the power distribution board (PDB). Four 30A ESCs distribute power to motors. A 5 V BEC on the PDB powers the Arduino Mega, Uno, and ESP32. A dedicated voltage divider ($R1=10k\Omega$, $R2=4.7k\Omega$) feeds the Mega ADC for battery monitoring. The MPU-6050 IMU and PCA9685 PWM driver operate at 3.3 V from their on-board regulators.

2.4 Propulsion System

Item	Specification	Notes
Motor	Brushless 2212 / 920 KV	All 4 positions
Propeller	8 × 4.5 inch (8045)	CW and CCW variants required
ESC	30A with BLHeli firmware	All 4
Battery	3S 2200 mAh LiPo, 30C	Single cell
Max thrust (single)	≈ 380 g at 100% throttle	At 11.1 V

3. Normal Procedures

3.1 Pre-Flight Checklist

MECHANICAL INSPECTION

- All 4 propellers tight and correctly oriented (CW on FL/RR, CCW on FR/RL)
- Frame arms locked — no cracks, no looseness
- Landing gear secure
- Motor shafts free-spinning — no binding or grating
- All screws torqued — vibration check complete

ELECTRICAL INSPECTION

- LiPo cell voltages balanced (> 3.9 V/cell, < 4.2 V/cell)
- All ESC connectors fully seated
- All signal wire connectors secure
- No damaged or bare wiring visible

FIRMWARE READY CHECK

- Power on Mega + Uno — serial monitors show READY within 5 s
- Wi-Fi SSID 'SUDARSHAN_AP' visible on GCS device

GCS PREFLIGHT TESTS

- GCS connected (green indicator in title bar)
- MOTOR_TEST — all 4 motors spin in correct sequence (PROPS OFF)
- IMU_CHECK — pass (green), roll/pitch/yaw $< \pm 2^\circ$ on level surface
- SONAR_CHECK — pass (green), alt_cm displayed > 0
- bat_mv $> 10\,500$ (≥ 3.5 V/cell) confirmed

AREA CLEARANCE

- All personnel ≥ 10 m from aircraft before ARM
- Wind assessed < 5 m/s
- Flight path clear of obstacles to planned altitude
- Observer designated if operating beyond 20 m

3.2 ESC Arming Sequence

ESCs arm automatically when the FC sends the minimum-throttle signal on power-on. The startup sequence of beeps confirms arming: one long beep = power connected; descending beeps = ESC armed. If an ESC does not beep, do NOT proceed — investigate before flight.

3.3 Takeoff

- Verify all pre-flight items checked.
- Ensure all personnel are ≥ 10 m clear.
- In GCS FLIGHT tab: issue ARM command. Confirm state = ARMED.
- Issue HOVER command. Drone will spin up and climb to hover altitude (~50 cm).
- Observe stabilization. Allow 5 s before commanding flight.

3.4 Normal Flight

Monitor telemetry continuously: alt_cm, bat_mv, sonar_ok, and dms_countdown. Keep dms_countdown > 5 s by sending periodic heartbeat (GCS does this automatically when connected). Keep bat_mv > 10 500 mV for comfortable margins.

3.5 Normal Landing

- Issue LAND command from GCS FLIGHT tab.
- Monitor state: HOVER → LAND → auto-DISARMED.
- Confirm motors stop after touchdown (sonar < 8 cm × 6 counts).
- Disconnect LiPo within 2 minutes of landing.

3.6 Post-Flight

- Disconnect LiPo — store at 3.8 V/cell (storage charge) if not flying again within 24 h.
- Inspect propellers for nicks, cracks, or balance issues.
- Check all connectors for heat or discoloration.
- Log flight duration, battery condition, and any anomalies in the flight log.

4. Emergency Procedures

All emergency procedures require immediate, decisive action. Do not hesitate. Safety of persons has absolute priority over equipment.

■ UNCOMMANDED YAW ROTATION

Issue LAND immediately.

If LAND does not take effect within 2 s, issue KILL.

After landing/KILL, power-cycle drone.

Root cause: wrong prop direction or IMU_YAW_SIGN. Inspect before next flight.

■ ALTITUDE LOSS / UNCONTROLLED DESCENT

Issue HOVER to re-engage altitude hold.

If descent continues, issue KILL immediately to prevent uncontrolled impact.

Root cause: sonar_ok = 0 or altitude PID fault. Check sonar before next flight.

■ GCS LOSS / TELEMETRY FREEZE

DMS will automatically issue LAND after 30 s of no heartbeat.

Visually monitor drone — it should descend and land autonomously.

If drone does not land within 40 s, use physical E-stop if fitted.

Do NOT approach aircraft until motors have stopped for ≥ 5 s.

■ BATTERY CRITICAL ($\text{bat_mv} \leq 9\,900\text{ mV}$)

FC automatically issues LAND — do not override.

Clear landing area immediately.

After landing, disconnect LiPo immediately — swollen cells are a fire hazard.

Do not recharge a deeply discharged or swollen LiPo.

■ MOTOR FAILURE (one motor stops)

Drone will lose control immediately — issue KILL.

Clear area — drone will fall.

Investigate cause: ESC fault, motor winding, prop strike.

Replace ESC/motor before next flight; rebalance and test-hover at low altitude.

■ KILL PROCEDURE

Issue KILL command from GCS FLIGHT tab.

All motors cut immediately — drone falls.

Use only when continued flight poses greater risk than fall impact.

After KILL, a full power-cycle of the drone is required before re-arming.

5. Systems & Indicators

5.1 Telemetry Field Definitions

Field	Description	Range / Notes
state	FC flight state	IDLE / ARMED / HOVER / LAND / KILL / OVERRIDE
roll_deg	Roll angle (°)	Positive = right wing down
pitch_deg	Pitch angle (°)	Positive = nose up
yaw_deg	Yaw angle (°)	0–360°, clockwise from north
alt_cm	Sonar altitude (cm)	0 = ground; valid range 5–200 cm
sonar_ok	Sonar validity flag	1 = valid; 0 = no echo / out of range
bat_mv	Battery voltage (mV)	Full = 12 600 mV; low = 9 900 mV
dms_countdown	DMS seconds remaining	Counts down from 30; reaches 0 → LAND
seq	GCS packet sequence number	Used to detect dropped packets
loop_ms	FC loop execution time (ms)	Nominal = 4 ms (250 Hz); > 6 ms = overload

5.2 DMS (Dead-Man Switch) Countdown

The DMS operates at three independent levels: (1) ESP32 heartbeat timer — resets on any TCP packet; (2) Mega FC internal timer — resets on any valid serial command; (3) GCS software watchdog — alerts operator if no telemetry for 5 s. If the FC timer reaches 0 (30 s), LAND is issued automatically. `dms_countdown` in telemetry reflects the FC-side timer.

5.3 Battery State Table

Voltage	Charge %	State	Indicator	Action
12.6 V (4.20 V/cell)	100%	Fully charged	WHITE	Normal operations
12.0 V (4.00 V/cell)	75%	Good	GREEN	Normal operations
11.4 V (3.80 V/cell)	50%	Adequate	GREEN	Plan landing soon
10.8 V (3.60 V/cell)	25%	Low	AMBER	Return and land
10.5 V (3.50 V/cell)	15%	Very Low	AMBER	Land immediately
9.9 V (3.30 V/cell)	< 5%	Critical	RED	Auto-LAND triggered
< 9.9 V	0%	Over-discharged	RED	Do not recharge — inspect

6. Performance Data

6.1 Endurance

Hover endurance on a fully-charged 2200 mAh 3S LiPo is approximately 8–12 minutes depending on payload and ambient conditions. Aggressive manoeuvres reduce endurance by up to 30%. Always land with a minimum of 2 minutes reserve (bat_mv $\approx$ 10 500 mV).

6.2 Speed

Mode	Speed	Notes
Hover speed	0 m/s	Altitude-hold hover
Slow cruise (GUIDE)	0.4 m/s	Minimum controlled translation
Fast cruise (GUIDE)	2.0 m/s	Maximum recommended speed
Max descent rate	0.5 m/s	LAND mode controlled descent
Max ascent rate	0.8 m/s	HOVER to target altitude

6.3 Altitude & Sonar Performance

Parameter	Value	Conditions
Altitude hold accuracy	± 5 cm	In still air, smooth ground surface
Sonar valid range	5 – 200 cm	HC-SR04 with 10° beam angle
Sonar update rate	10 Hz	Triggered by FC loop
Altitude response time	< 0.3 s	Step response to ± 10 cm disturbance

6.4 Control & PID Parameters (Default)

Axis	Kp	Ki	Kd	Loop Rate
Roll	1.8	0.02	0.6	250 Hz
Pitch	1.8	0.02	0.6	250 Hz
Yaw	2.5	0.05	0.0	250 Hz
Altitude	3.0	0.08	1.2	250 Hz

End of SUDARSHAN UAV Operator Manual — Issue 1.0